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ISSUE NO. ____________

TITLE: INSTALLATION, OPERATION AND MAINTENANCE MANUAL FOR MODEL LTL-4 GRADIENT WATER HEAT EXCHANGER

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REV.	PAGE	CHANGE RECORD	PREP. BY	ENG. APPVL.	Q.A. APPVL.
0	All	Original document released. (See cover sheet)			
1	5,7 6-12	Add information on power cable Add French and Spanish translations of warnings	BAA 01/26/99	KDD 01/26/99	RRS 01/26/99
2	5,7 9 10 11	Add GEMS PDU as exclusive power source Add reset procedure for time-delay relay Add low flow and cabinet access interlocks Add over-ride of cabinet access interlock	BAA 02/09/99	KDD 02/09/99	RJR 02/09/99
3	1 3 4,5,7, 8,9,11 12,14,19	Add G.E. part number. Revise Table of Contents. Add SI Units.	BAA 05/26/99	KDD 05/26/99	TJK 05/26/99
	4	Revise cooling capacity (Paragraph I). Add factory set-point temperature. (Para. I). Add pump dead-head pressure (Paragraph I). Add wet and shipping weights (Paragraph I).			
	5	Revise maximum power requirement (Para. I). Add features (Paragraph I). Add min. facility supply pressure (Para. I). Revise input voltage (Paragraph I). Add electrical connector (Paragraph I).			
	6	Add statement on installation of cabinet top (Paragraph 2-1). Add warning about magnetic field (Para. 2-1). Add installation requirements (Para. 2-2).			
	7	Add warning about ethylene glycol (Paragraph 2-3).			
	8	Add regulations compliance statement (Paragraph 2-3). Revise fluid connection labels (Para. 2-3).			
	8,11	Add statement on switches (Para. 2-4 & 3-2).			
	11	Revise flow capacity, thermal capacity, maximum ambient temperature (Para. 3-3).			
	11,12,13	Add description to electrical and control design and components (Para. 3-3).			
	13	Revise schedule for coolant level preventative maintenance (Para. 4-2). Add schedule for strainer preventative maintenance (Para. 4-2).			
	14	Add setpoints for controls and interlocks (Paragraph 4-4).			
	15	Expand troubleshooting guidelines (Para 4-5). Revise replacement parts list (Para. 4-6).			

REV.	PAGE	CHANGE RECORD	PREP. BY	ENG. APPVL.	Q.A. APPVL.
4	4,7,9 17,23 6 9 10 15 16 21 22	Add reference to Unit Assembly Drawing, T2583-0201. Add specifications for fuse for control transformer. Add to description of removal of cabinet top. Add location of air purge hose assembly. Add procedure for setting temperature controller and control valve. Correct temperature controller output voltage. Correct transformer part number. Add fuse block and fuse.	BAA 6/8/99	KDD 6/8/99	SEC 6/8/99
5	4,7,9 17,23 4,9 11,12,13 15,18,19 20,22,23 16 22	Change drawing number of Unit Assembly Drawing. Change drawing number of Wiring Schematic. Correct control voltage for temperature controller and flow control valve. Change part numbers and quantites of quick disconnect assemblies.	BAA 11/16/99	KDD 11/16/99	RRS 11/16/99
6	App.	Added drawings to the document	CAR 12/17/03	GAZ 12/17/03	RRS 2/17/03
7	1	Added Proprietary Statement to cover page	CAR 2/10/05	TJH 2/10/05	ERS 2/10/05
8	5 4-22 9, 12	Added power cord length requirement Corrected mis-numbered pages Added power switch to procedure	CAR 3/9/07	JKS 3/9/07	RRS 3/9/07
9	17,18	Added 4-7 Removal Process for Mobiles	JAD 9/20/07	TJH 9/20/07	KAH 9/20/07

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I. TABLE OF PARTICULARS

PERFORMANCE	
COOLING CAPACITY	4.5 KW, MAX.
GRADIENT COIL COOLANT SUPPLY TEMPERATURE FACTORY SETPOINT	55-77°F (13-25°C) 64°F (18°C)
FREQUENCY	50-60 HZ
COIL COOLANT FLOW RATE	3.0 GPM (11 L/MIN.) MAX. @ 50 HZ, 3.5 GPM (13L/MIN.) MAX. @ 60 HZ MAX.
COIL COOLANT OUTLET PRESSURE	20-60 PSIG (140-410 KPA) NORMAL OPERATION 75 PSIG (520 KPA), MAX., PUMP DEAD-HEAD
RESERVOIR CAPACITY	1 GALLON (3.8 L)
OPERATING AMBIENT	50-113°F (10-45°C)
COIL COOLANT	35% ETHYLENE GLYCOL/65% DEIONIZED WATER BY VOLUME
CHILLED FACILITY COOLANT	50% ETHYLENE GLYCOL/50% DEIONIZED WATER BY WEIGHT
TEMPERATURE CONTROL	FACILITY COOLANT THROTTLING

MECHANICAL	
PUMP	CENTRIFUGAL, SINGLE STAGE, PLASTIC
COIL COOLANT/CHILLED FACILITY COOLANT HEAT EXCHANGER	STAINLESS STEEL, BRAZED PLATE
COIL COOLANT CIRCUIT MATERIAL	PLASTIC, STAINLESS STEEL, COPPER
CHILLED FACILITY COOLANT CIRCUIT MATERIAL	PLASTIC, STAINLESS STEEL, COPPER
PUMP INLET STRAINER	STAINLESS STEEL, 40 MESH
RESERVOIR MATERIAL	POLYETHYLENE
HOUSING AND BRACKETS	ALUMINUM
METAL FINISH	POWDER COAT
TEMPERATURE CONTROL VALVE	2-WAY BALL
WEIGHT	86 LBS (39 KGS) (DRY, NET SHIPPING) 98 LBS. (44 KGS) (WET) 140 LBS (64 KGS) (ESTIMATED GROSS SHIPPING)
SIZE	15" (380 MM) W X 17" (430 MM) L X 19" (480 MM) H WITH ISOLATOR FEET

ELECTRICAL	
POWER SOURCE (EXCLUSIVE)	PDU IN GEMS MRI SYSTEM, DEDICATED CIRCUIT BREAKER - 110/115VAC, 15A, 1PH, 50-60Hz
MAXIMUM POWER	1000 VA
PUMP MOTOR	1.0 HP (750 W)
MOTOR PROTECTION	THERMAL, 250°F (120°C) MAX. TRIP
POWER CABLE (BY OTHERS)	3 CONDUCTOR, 12 AWG, TYPE SO, FT4 FLAMMABILITY, WITH A 3-PIN, TWIST-LOCK, SOCKET CONNECTOR, MINIMUM LENGTH OF 4 FT., MAXIMUM LENGTH OF 10 FT.
FUSE, CONTROL TRANSFORMER, PRIMARY SIDE, "HIGH" LEAD	1 AMP, 125 VAC OR GREATER, "SLOW-BLOW"-APPROX. 1 SECOND AT 3.5 AMPS

FEATURES
COOLANT SUPPLY TEMPERATURE CONTROL
COOLANT SUPPLY TEMPERATURE READOUT
LOW COOLANT LEVEL INTERLOCK
LOW FLOW COOLANT INTERLOCK
HIGH-TEMPERATURE COOLANT INTERLOCK
CABINET SAFETY INTERLOCK
POWER INDICATOR LIGHT

SYSTEM REQUIREMENTS SUPPLIED TO THE LTL-4	
FACILITY CHILLED COOLANT:	
COMPOSITION	50±5% ETHYLENE GLYCOL/50±5% WATER BY WEIGHT
FLOW RATE	3.3 ± 0.3 GPM (12 ± 1 L/MIN.)
PRESSURE DROP INTERNAL TO LTL-4	8 PSID @ 3.0 GPM (55 KPA @ 11 L/MIN.)
INLET TEMPERATURE	52°F (11°C) , MAX.
INLET PRESSURE	20 PSIG (140 KPA), MIN., 50 PSIG (340 KPA), MAX.
FLUID CONNECTIONS	BARBS FOR 1/2" (13 MM) ID HOSE
POWER:	
VOLTAGE	110 VAC ± 10%

SYSTEM REQUIREMENTS SUPPLIED TO THE LTL-4	
CURRENT	12.5 A STEADY STATE AT 110 VAC, 60 HZ 75 A, IN-RUSH AT 110 VAC, 60 HZ
FREQUENCY	50 OR 60 HZ
PHASE	1
CONNECTION ON CABINET	HUBBELL 4716 C

II. INSTALLATION

2-1 PREOPERATIONAL CHECK

No special unpacking or handling procedures are required. Normal careful handling will prevent marring of the finish or damage to the electrical and fluid connections. Remove cabinet top and inspect system for dirt and loose wires. Inspect all components for dirt and loose packing materials. Refer to Ellis & Watts (E&W) Unit Assembly Drawing, T2583-0201-01. Check particularly the temperature controller and the pump/motor mount. Dirt and foreign matter can cause electrical shorting in the controller.

Check all electrical connections for loose wires. All wires shall be inspected for frayed or worn insulation. If any wires are found defective, they shall be replaced.

The cabinet top must be installed on the Model LTL-4 Gradient Water Heat Exchanger before it is delivered to the customer.

WARNING

RISK OF ELECTRIC SHOCK. CAN CAUSE INJURY OR DEATH. DISCONNECT ALL REMOTE ELECTRIC POWER SUPPLIES BEFORE SERVICING.

AVERTISSEMENT

RISQUE DE CHOC ELECTRIQUE POUVANT PROVOQUER DES BLESSURES OU LA MORT. DEBRANCHER TOUTES LES ALIMENTATIONS ELECTRIQUES EXTERNES AVANT L'ENTRETIEN.

ADVERTENCIA

RIESGO DE CHOQUE ELECTRICO. PUEDE CAUSAR LESIONES O LA MUERTE; DESCONECTAR TODO SUMINISTRO REMOTO DE ENERGIA ELECTRICA ANTES DE REALIZAR LABORES DE MANTENIMIENTO.

WARNING

CONTAINS FERROUS MATERIAL. PERSONAL INJURY COULD RESULT DUE TO STRONG ATTRACTION TO MAGNET.

AVERTISSEMENT

CONTIENT UNE MATIÉRE FERREUSE. UNE BLESSURE CORPORELLE POURRAIT ÊTRE PROVOQUÉE PAR LA FORTE ATTRACTION MAGNÉTIQUE.

ADVERTENCIA

CONTIENE MATERIAL FERROSO. LA FUERTO ATRACCIÒN AL IMÁN PODRIA CAUSAR LESIONES PERSONALES.

2-2 INSTALLATION

The Model LTL-4 Gradient Water Heat Exchanger (GWHX) is designed to sit on its four, vibration-dampening, support feet. It should not be suspended by its top cover on vertical sides. The GWHX must be prevented from moving or tipping; one method is to band it to a vertical wall.

Ventilation air flow must not be restricted from passing through the cabinet louvers. One set of louvers on one side of the cabinet must be completely unblocked with no obstructions within 18 inches (460 mm) from the louvered side. The opposite set of louvers may be positioned against a SOLID vertical wall; however the louvers should not be placed against a soft (e.g., carpeted) vertical wall that might conform to the shape of the louvers and totally block air flow.

2-3 CONNECTIONS

WARNING

CHILLED COOLANT MUST BE SUPPLIED TO THE LTL-4 AT ALL TIMES THE GRADIENT COOLING WATER COIL PUMP IS OPERATING TO INSURE THAT THE PLASTIC HOSES ARE NOT OVER-HEATED BY PUMP HEAT OR HEAT FROM THE GRADIENT COIL.

AVERTISSEMENT

DU LIQUIDE DE REFROIDISSEMENT REFROIDI DOIT ETRE FOURNI A LT-4 EN TOUT TEMPS LORSQUE LA POMPE A BOBINE DE GRADATION D'EAU DE REFOIDISSEMENT FONCTIONNE, AFIN D'ASSURER QUE LES TUYAUX DE PLASTIQUE NE SOIENT PAS

SURCHAUFFES PAR LA CHALEUR DE LA POMPE OU PAR LA CHALEUR PROVENANT DE LA BOBINE DE GRADATION.

ADVERTENCIA

SE DEBERA SUMINISTRAR REFRIGERANTE ENFRIADO AL LTL-4 SIEMPRE QUE LA BOMBA DEL SERPENTIN DE AGUA PARA ENFRIAMIENTO ESCALONADO ESTE FUNCIONANDO PARA ASEGURARSE QUE LAS MANGUERAS PLASTICAS NO SE SOBRECALIENTEN DEBIDO AL CALOR DE LA BOMBA O AL CALOR DEL SERPENTIN DE GRADIENTE.

WARNING

CONTAINS ETHYLENE GLYCOL. HARMFUL OR FATAL IF SWALLOWED. CAUSES EYE IRRITATION. PROLONGED OR REPEATED BREATHING OF MIST OR VAPOR HARMFUL.

AVERTISSEMENT

CONTIENT DE L'ÉTHYLÉNEGLYCOL. NOCIF OU FATAL EN CAS D'ABSORPTION. PROVOQUE L'IRRITATION OCULAIRE. UNE INHALATION PROLONGÉE OU RÉPÉTÉE DE LA BUÉE OU VAPEUR EST DOMMAGEABLE.

ADVERTENCIA

CONTIENE ETILENGLICOL. NOCIVO O FATAL SI SE INGIERE. PRODUCE IRRITACIÒN DE LOS OJOS. LA INHALACIÒN PROLONGADA O REPETIDA DE VAHO O VAPOR ES NOCIVA.

Position the GWHX in its permanent location. The Model LTL-4 GWHX shall be used only as part of the General Electric Medical Systems (GEMS) Magnetic Resonance Imaging System. The unit shall be powered through the GEMS Power Distribution Unit (PDU). It is powered from a transformer secondary side; a dedicated, 15A, circuit breaker is used for the GWHX. All electrical connections must comply with all regulations in force at the location of operation.

Fluid line connections must be made to the GWHX inlet and outlet ports before the unit is filled. The chilled facility coolant inlet and outlet ports are 1/2" (13 mm) hose barb connections, labeled as "FACILITY CHILLED WATER RETURN" and "SUPPLY", respectively. Facility coolant is provided to the GWHX through the "RETURN" connection. The gradient cooling water coil (GCWC) connections are 1/2" (13 mm) hose barb connections, labelled as "COIL SUPPLY" and "RETURN". (See Ellis & Watts [E&W] Interface Control Drawing T2583-0101.) Check that all fluid lines in the complete system are tight. Any leakage will cause loss of pressure which will lower effectiveness of the system and cause failure.

Electrical power is supplied through the three-pin, twist-lock, socket connector. The power cable is a three-conductor, 12 AWG, Type SO cable with a FT4 flammability rating. The cable is supplied by others and must comply with all regulations in force at the location of operation and as listed in Electrical Requirements Table on page 6. Use the grounding stud to properly ground the GWHX. ON/OFF switching of the unit must be provided by the user external to the GWHX. (Refer to the Interface Control Drawing and the Wiring Schematic, E&W Drawings T2583-0101 and T2583-1202-01, respectively.)

2-4 INITIAL FILLING

This procedure outlines the steps for initially filling the Gradient Cooling Water Coil (GCWC) loop of the Gradient Water Heat Exchanger (GWHX). The procedure applies regardless of whether or not the gradient coil already contains coolant. Refer to Section 3-3 regarding the operation of the level switch, high-temperature limit thermostat, cabinet interlock switch and flow-switch. These interlock switches must be closed to operate the pump. Refer to E&W Unit Assembly Drawing, T2583-0201-01, for location of components. Note particularly that removing the cabinet top interrupts operation by opening the cabinet interlock switch. During filling and other maintenance, the switch may be overridden by depressing the switch plunger. This may be accomplished using a weight, c-clamp, etc.

1. Remove the entire cabinet top, by removing four Phillips-head screws in each of two sides of the unit (eight screws, total).

   Refer to Drawing T2583-0201-01.

2. Insure that the GCWC loop and the chilled facility coolant supply loop external to the GWHX contain no leaks or open vents.

3. Supply chilled coolant to the GCWC, insuring all air is purged from the lines.

WARNING

CHILLED COOLANT MUST BE SUPPLIED TO THE LTL-4 AT ALL TIMES THAT THE GCWC PUMP IS OPERATING TO INSURE THAT THE PLASTIC HOSES ARE NOT OVER-HEATED BY PUMP HEAT OR HEAT FROM THE GRADIENT COIL.

AVERTISSEMENT

DU LIQUIDE DE REFROIDISSEMENT REFROIDI DOIT ETRE FOURNI A LT-4 EN TOUT TEMPS LORSQUE LA POMPE A BOBINE DE GRADATION D'EAU DE REFOIDISSEMENT FONCTIONNE, AFIN D'ASSURER QUE LES TUYAUX DE PLASTIQUE NE SOIENT PAS

SURCHAUFFES PAR LA CHALEUR DE LA POMPE OU PAR LA CHALEUR PROVENANT DE LA BOBINE DE GRADATION.

ADVERTENCIA

SE DEBERA SUMINISTRAR REFRIGERANTE ENFRIADO AL LTL-4 SIEMPRE QUE LA BOMBA DEL SERPENTIN DE AGUA PARA ENFRIAMIENTO ESCALONADO ESTE FUNCIONANDO PARA ASEGURARSE QUE LAS MANGUERAS PLASTICAS NO SE SOBRECALIENTEN DEBIDO AL CALOR DE LA BOMBA O AL CALOR DEL SERPENTIN DE GRADIENTE.

4. Fill the GWHX reservoir to within 1.5 to 2 inches (38 to 51 mm) from its top, so that the level in the reservoir is above the pump discharge connection.

5. Connect the air purge hose assembly supplied with the GWHX to the access valve at the pump discharge. Secure the opposite end of the purge hose in the reservoir. The air purge hose assembly is stored inside the cabinet, attached using a Velcro strap.

6. Start the GCWC pump by applying power to the unit.

7. Observe the pump suction hose between the reservoir connection and the pump suction connection. If a large air slug(s) is observed in the hose and the coolant does not appear to be flowing, disconnect power to the GWHX (pump), immediately.

WARNING

EXTENDED OPERATION OF THE PUMP WITHOUT PROPER COOLANT FLOW TO THE PUMP MAY DAMAGE THE PUMP.

AVERTISSEMENT

LE FONCTIONNEMENT PROLONGE DE LA POMPE SANS CIRCULATION ADEQUATE DE LIQUIDE DE REFROIDISSEMENT EN DIRECTION DE LA POMPE PEUT ENDOMMAGER LA POMPE.

ADVERTENCIA

LA OPERACION PROLONGADA DE LA BOMBA SIN EL FLUJO ADECUADO DE REFRIGERANTE HACIA LA BOMBA PODRIA DANAR LA BOMBA.

8. Observe the coolant level in the reservoir. If necessary, add coolant to maintain the level above the pump discharge connection. However, do not fill the reservoir above 1.5 to 2 inches (38 to 51 mm) below the top of the reservoir.

9. Allow time for any air slugs in the suction hose to migrate into the pump, so the suction hose is completely filled with coolant.

10. After an additional 10 seconds, re-apply power to operate the pump.

11. Repeat Steps 7 through 10 to purge the air from the GCWC loop through the air purge hose. Depending on the volume of the loop, ten to fifty iterations may be required. Some coolant will be pumped through the air purge hose along with the air slugs into the reservoir. Note that the coolant level in the reservoir drops as air is purged from the loop.

12. After noticeable air slugs have been purged from the loop, operate the pump for an additional 10 minutes with the air purge hose connected.

13. Shut-off power to the GWHX.

14. Remove the air purge hose assembly. Install the access valve cap and the reservoir cap.

15. Install the cabinet top.

16. Re-start the GWHX and observe the coolant level in the reservoir. If necessary, add or remove coolant through the access panel in the housing top until the final coolant level is approximately at the midpoint between the "ADD" and "FULL" lines.

III. OPERATION

3-1 STARTING

To start the GWHX, simply make the proper power connection to the power socket and turn the power switch on. Refer to E&W Drawing T2583-1202-01, Wiring Schematic.

If the GWHX is automatically stopped by the built-in interlock for coil coolant low-flow, a time-delay relay must be reset before the unit can be re-started. The reset is accomplished by: 1) removing and re-installing the cabinet top to exercise the cabinet access interlock switch, or 2) un-plugging and plugging-in the electrical inlet plug.

Upon initial start-up, use a properly calibrated flow meter to check the fluid flow rate of the GCWC coolant and the chilled facility coolant. Note that the flow-rate of the chilled coolant will be restricted by the two-way control valve if full capacity is not being called-for.

WARNING

THE PUMP IS A CENTRIFUGAL PUMP. IF THE OUTLET IS EXCESSIVELY RESTRICTED, THE OUTLET PRESSURE CAN RISE TO DAMAGING PRESSURE.

AVERTISSEMENT

CETTE UNITE EST UNE POMPE CENTRIFUGE. SI LA SORTIE EST EXCESSIVEMENT OBSTRUEE, LA PRESSION DE SORTIE PEUT S'ELEVER A UN NIVEAU DOMMAGEABLE.

ADVERTENCIA

LA BOMBA ES UNA BOMBA CENTRIFUGA. SI SE RESTRINGE LA DESCARGA DE MANERA EXCESIVA, LA PRESION DE DESCARGA PUEDE ELEVARSE HASTA UN NIVEL DANINO.

3-2 STOPPING

The GWHX is stopped by either disconnecting the power cable at the socket connector or by shutting-off the power remotely. If the cabinet top is removed, the safety interlock switch will also stop the unit.

3-3 THEORY OF OPERATION

The GWHX is a system designed as a liquid-to-liquid heat transfer system. The system consists of a supply system for ethylene glycol coolant to the GCWC, reservoir and a heat exchanger cooled by a chilled ethylene glycol/water mixture. The system circulates a minimum of 3 GPM (11 l/min) of GCWC coolant at a maximum system outlet pressure of 60 PSIG (410 KPa) at 60 Hz and at a minimum outlet temperature of 55°F (13°C) while operating in an ambient temperature up to 113°F (45°C). The GCWC coolant is circulated through a gradient coil, where it absorbs a maximum of 15,400 Btu/hr (4.5 kw), generated by the equipment. The fluid then returns to the heat exchanger, where heat is transferred to the chilled facility coolant.

The fluid circuits are shown schematically in the E&W Piping Schematic, Drawing T2583-1201. Coolant returning from the gradient coil enters the GWHX through the "COIL RETURN" connection. A standpipe-type reservoir is included to accommodate thermal expansion, provide a readily-accessible fill port and enable purging of air from the system. The pump takes coolant, by suction, from the "RETURN" connection and discharges into the heat exchanger through a 40-mesh strainer. Hot gradient coolant is therefore supplied from the pump to the heat exchanger and cold gradient water is returned from the heat exchanger. Heat is transferred in the heat exchanger to the chilled facility coolant flowing through the heat exchanger.

The electrical design is shown in the E&W Wiring Schematic, Drawing T2583-1202-01. Power (110 VAC ± 10%/1 phase/50-60 Hz/12.5 A) is supplied via a three-pin, twist-lock, socket connector. Power is distributed through a terminal block. Full voltage is supplied to the pump motor, through a contactor (closed when the coil is energized). The contactor coil is wired in series with a level switch, located in the reservoir, a high-temperature limit thermostat, a cabinet access interlock and a coil coolant flow switch. Thus, the pump stops if the pump or pump motor overheats, a low-level or low flow condition occurs or the cabinet top is removed. A 110 VAC/24 VAC transformer supplies power to the control circuit for the temperature controller, temperature display and two-way ball valve actuator.

The temperature of the coolant supplied to the GCWC is controlled by modulating the chilled facility coolant flow rate. A positive temperature coefficient, silicon sensor senses the actual coolant supply temperature. The operator dials-in the desired coolant supply temperature. The temperature control module compares the set-point and actual coolant supply temperatures. Using proportional plus integral logic, the controller supplies a 2-10 VDC signal to the actuator on the two-way, control, ball valve. The two-way valve adjusts to pass more chilled coolant through the heat exchanger if the actual temperature is greater than the set-point temperature and less chilled coolant through the heat exchanger if the actual temperature is less than the set-point temperature. The actual temperature is shown continuously on the temperature display module. A push button enables the set-point temperature to be displayed. Voltage and power requirements for the controller, display and actuator are shown on the Wiring Schematic, Drawing T2583-1202-01.

The green neon, power indicator lamp (IL) on the side of the cabinet is lighted when voltage is applied to the line-side contacts of the motor contactor (K1), to the primary side of the control transformer (T1) and across the following components which are wired in series: reservoir level switch (LS), high-temperature limit thermostat (TH), cabinet interlock switch (IS), flow switch (FS) and coil of the motor contactor (K1).

The level switch (LS) is located in the reservoir (R) for the coil coolant. It is positioned at approximately the one-third full level. If the coolant level drops below this level (for example, due to a leak in the coil coolant circuit), the level switch opens. This interrupts power to the coil of the motor contactor (K1), opens the contacts and stops the pump motor (P/M) to prevent the pump from overheating.

The high-temperature limit thermostat (TH) is fastened to the copper tee at the outlet of the brazed-plate heat exchanger (HX) for the coil coolant. The thermostat protects the pump (P/M) from overheating by opening if the coil coolant temperature rises above 100°F (38°C). This would occur if facility coolant flow is interrupted or its temperature increases significantly from approximately 52°F (11°C) to 85°F (29°C). The thermostat closes when cooled below 80°F (27°C) to enable the unit to again operate.

The cabinet interlock switch (IS) is a safety feature for personnel. When the cabinet top is removed, the switch opens and interrupts power to the coil of the pump motor contactor (K1). This causes the pumps (P/M) to stop until the cabinet top is replaced. During servicing, the switch may be over-ridden by depressing the switch plunger using a weight, c-clamp, etc.

The flow switch (FS) is located in the coil coolant circuit, upstream of the pump (P/M) and reservoir (R). Its purpose is to protect the pump from overheating if coil coolant is interrupted (for example, by a leak in the coil coolant circuit). If coolant flow decreases below one gpm (3.8 l/min), the switch opens to interrupt power to the coil of the motor contactor (K1). This opens the contacts and stops the pump motor (P/M). To enable the pump to operate at initial start-up when no coil coolant is flowing, a time delay relay (TDR) is wired in parallel with the flow switch. It provides a closed electrical circuit "around" the flow switch until the pump can operate for a sufficient time to establish coolant flow greater than one gpm (3.8 l/min.). The relay is adjustable and is factory-set to remain closed for about 30 seconds. After 30 seconds, coolant flow is established through the flow switch to close it, the time delay relay opens and the pump continues to operate. If the flow switch opens at any time and stops operation of the unit, the time delay relay must be reset by cycling power to it, before operation can be resumed. Power to the relay can be cycled by any one of the following methods:

1) Dis-connect and re-connect the main power plug (P/S).

2) Removing and installing the cabinet top, i.e., cycle the cabinet interlock switch (IS).

IV. MAINTENANCE

WARNING

RISK OF ELECTRIC SHOCK. CAN CAUSE INJURY OR DEATH. DISCONNECT ALL REMOTE ELECTRIC POWER SUPPLIES BEFORE SERVICING.

AVERTISSEMENT

RISQUE DE CHOC ELECTRIQUE POUVANT PROVOQUER DES BLESSURES OU LA MORT. DEBRANCHER TOUTES LES ALIMENTATIONS ELECTRIQUES EXTERNES AVANT L'ENTRETIEN.

ADVERTENCIA

RIESGO DE CHOQUE ELECTRICO. PUEDE CAUSAR LESIONES O LA MUERTE; DESCONECTAR TODO SUMINISTRO REMOTO DE ENERGIA ELECTRICA ANTES DE REALIZAR LABORES DE MANTENIMIENTO.

4-1 GENERAL

The GWHX is designed to require a minimum amount of maintenance. To insure that the system is ready for operation at all times, it should be inspected periodically and systematically so that potential problems may be discovered and corrected before they cause serious damage or failure. Problems discovered during operation of the system shall be noted for future correction, to be made as soon as operation is ceased. Stop operation immediately if a deficiency is noted which would damage the equipment if operation were continued.

During maintenance, with the cabinet top removed, the cabinet access interlock switch may be overridden to enable the equipment to operate. Operation is achieved by depressing the switch plunger using a weight, c-clamp, etc.

4-2 PREVENTATIVE MAINTENANCE

The frequency of preventative, periodic checks may be determined based on actual experience in the operating environment.

The coolant level in the reservoir should be checked at three-month intervals. The strainer for the coil coolant should be checked for cleanliness at three-month intervals during normal operation. If the coolant loop is opened or the possibility of coolant contamination exists, the unit should be operated for one hour and the strainer should then be checked. No lubrication is required on this system.

4-3 SPECIAL TOOLS

No special tools are required. Normal tools kept by experienced heat exchanger equipment mechanics are sufficient.

4-4 CONTROL AND INTERLOCK SETPOINTS

The setpoints given in Table IV-1 are provided for the control components and interlocks to aid in troubleshooting the unit. In addition, test point voltages are shown on the Ellis & Watts Wiring Schematic, Drawing T2583-1202-01.

For the temperature controller, the temperature setpoint is adjusted via the labeled rotary knob located on the outside face of the controller. The throttle range, operation mode jumpers, integral control DIP switches and minimum output voltage are set inside the controller. The inside can be accessed by removing the front of the controller, by removing one Phillips-head screw in each of the four corners of the controller front face.

The throttle range is set using the rotary knob labelled "THROT RANGE." The operation mode jumpers are set at Position J1 on the circuit board. For direct action, each of the two jumpers should be oriented horizontally (across the circuit board). If necessary, pull the jumpers directly out of the circuit board, rotate 90° in either direction and re-insert into the circuit board. The DIP switches are four labeled slide switches, clustered together on the circuit board. The minimum output voltage is set using the rotary knob labeled "MIN OUTPUT." At the main terminal board (see E&W Drawing T2583-1202-01), connect a voltmeter across "V" and "C". Then, rotate the knob until 2 VDC is displayed at the voltmeter.

For the control valve for facility water, rotation position and range are set on the outside face of the actuator cover, using a flat-head screw driver. Rotation position refers to the direction of valve rotation with increasing control voltage supplied to the actuator by the temperature controller. Its setting should be "L". Rotation range limits the amount of valve travel and should be set at "100%".

TABLE IV-1
CONTROL AND INTERLOCK SETPOINTS

COMPONENT AND PARAMETER	SETPOINT
Temperature Controller	
Temperature Setpoint	18°C (64°F)
Throttle Range	13°F (7°C)
Operation Mode Jumpers	Direct Action
Integral Control DIP Switches	
1	Off
2	Off
3	Off
4	On
Minimum Output Voltage	0 VDC
Flow Control Valve (Facility Water)	
Range	
Closed (0% Cooling)	0 VDC
Open (100% Cooling)	10 VDC
Rotation	
Position	L
Range	100%
Time Delay Relay	
Delay Time	30 sec.
Flow Switch	
Flow Required For Closed Switch	> 1 gpm (3.8 l/min.)
Thermostat, High-Temperature Limit	
Open On Temperature Rise	100±7°F (38±4°C)
Close on Temperature Fall	80±7°F (27±4°C)
Cabinet Interlock Switch	
Closed	Plunger Depressed
Reservoir Level Switch	
Closed	Float Horizontal
Pump Motor	
Thermal Overload Trip Temperature	250°F (120°C), Max.
Pump	
Capacity	3 gpm @ 60 psig @60 Hz (11 l/min. @ 410 KPa @ 60 Hz), 2.5 gpm @ 40 psig @ 50 Hz (9.5 l/min. @ 280 KPa @ 50 Hz)
Dead-head Pressure	75 psig (520 KPa)

4-5 TROUBLESHOOTING

A troubleshooting guide is provided in Table IV-2 to aid in identifying and correcting problems.

4-6 REPAIR AND REPLACEMENT PARTS

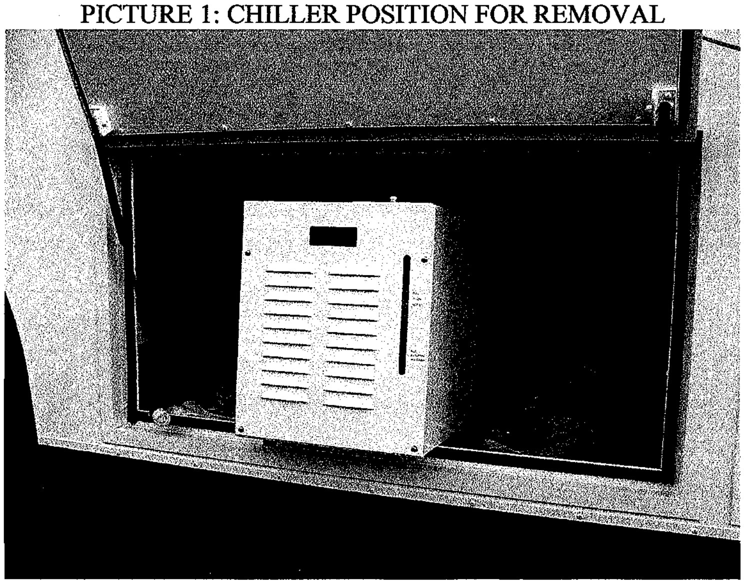
Due to the simplicity of construction of the GWHX, all of the major components are considered replaceable rather than repairable. Refer to Table IV-3 for the major components of the heat exchanger which may require replacement. Refer to the Unit Assembly Drawing, T2583-0201-01, for the location of these components. Ordinarily, replacement of a component shall take place only when found faulty through the use of the troubleshooting procedures found in Table IV-2.

4-7 REMOVAL PROCESS FOR MOBILES

The LTL4 chiller weighs 84 lbs (40kgs). The Hi-Jack Lifting Fixture must be used if one person is to replace the LTL4 chiller. The Hi-Jack Lifting Fixture is available from the GE Tools Depot. There is no part number for this device, it is known as the Hi-Jack, model A1. Alternative method to using the Hi-Jack is to use 2 persons to remove the LTL4 chiller from the mobile belly compartment.

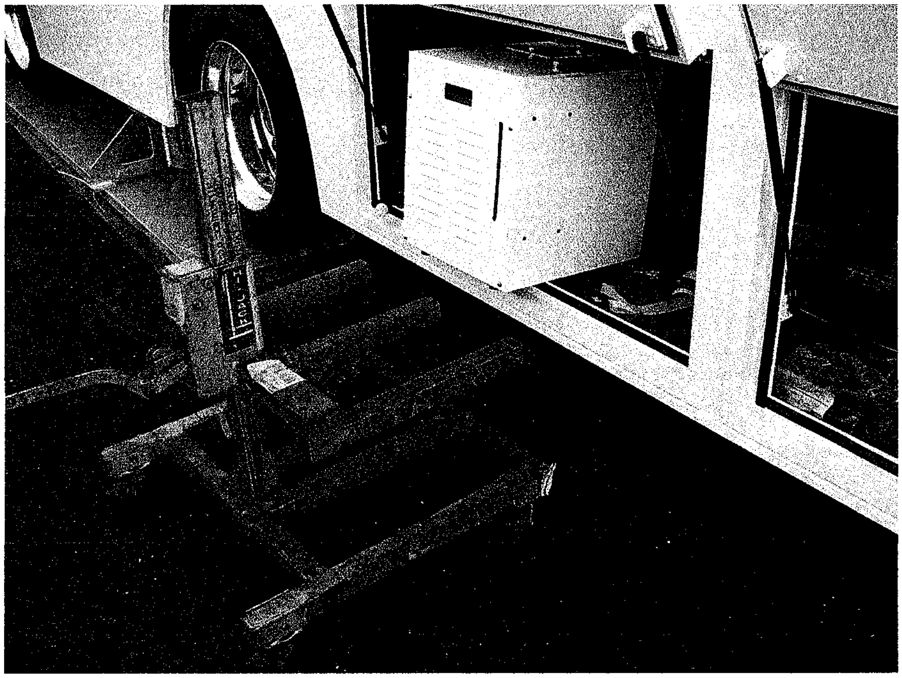
1) Turn off power to the LTL4 chiller at the PDU.
2) Turn off power to the Main Mobile Chiller.
3) There are two shut-off valves that are in series with the LTL4 chiller Input and Output from the Main Mobile Chiller. Turn the two valves to the "closed" position.
4) Remove the power cord from the LTL4 Chiller
5) Remove four waterlines from the LTL4 Chiller.
6) With the LTL4 chiller free, position the chiller to be removed. See Picture 1.

PICTURE 1: CHILLER POSITION FOR REMOVAL



7) If the Hi-Jack Lifting Tool is not available, use 2 persons to remove the LTL4 Chiller from the Mobile belly compartment.
8) Position the Hi-Jack lifting fixture near the LTL4 Chiller. See Picture 2.

PICTURE 2: POSITIONING OF HI-JACK



9) Carefully move the LTL4 Chiller from the Belly compartment of the Mobile onto the Hi-Jack.
10) When the LTL4 Chiller is completely removed from the Belly compartment, lower the Hi-Jack as this will increase stability when moving the Hi-Jack.
11) Remove failed LTL4 Chiller from the Hi-Jack, and place the new LTL4 Chiller onto the Hi-Jack.
12) Re-install the replacement LTL4 chiller into the mobile by reversing the steps in the mentioned procedure.

TABLE IV-2
TROUBLESHOOTING

TROUBLE	PROBABLE CAUSE	REMEDY
1. Low GCWC coolant flow	a. Low coolant supply	a. Check coolant level in reservoir. Refill, if necessary. Check Probable Cause c.
	b. Constriction in system	b. Check all coolant lines and heat exchanger for blockage and remove.
	c. Leak in system	c. Check all coolant lines for leaks and repair.
	d. Pump/motor malfunction	d. Check pump and motor and replace, if necessary.
	e. Low voltage or frequency f. No power to pump motor	e. Check power and wiring and repair. f. Check operation of contacts in motor contactor. Replace contactor, if necessary. Check for power to coil in motor contactor. If no voltage, check operation of reservoir level switch, high-temperature thermostat, cabinet access switch, flow switch and time delay relay. Electrically isolate each component and verify its open/close operation. Refer to Wiring Schematic, E&W Drawing T2583-1202-01. Replace, if necessary.
2. High GCWC coolant temperature	a. Low coolant flow	a. See Trouble Item 1.
	b. Low chilled facility coolant flow	b. Check and repair lines for leaks, lines for blockage, operation of two-way control valve, operation of chilled coolant pump.
	c. High facility coolant temperature d. Temperature control system malfunction	c. Check facility coolant return temperature. Repair chiller, if necessary. d. Check controller, sensor and valve actuator for correct settings. Refer to Table IV-1. Check components for correct input voltage. Refer to Wiring Schematic E&W Drawing T2583-1202-01. Vary temperature setpoint at controller while monitoring control signal at actuator and observing motion of valve. If control signal does not change or is

TROUBLE	PROBABLE CAUSE	REMEDY
		erratic, replace controller. If actuator does not respond to correct control signal, replace actuator.
3. No flow from pump	a. No power to unit	a. Check power and wiring; repair or replace.
	b. No power to pump motor	b. Check operation of contacts in motor contactor. Replace contactor, if necessary. Check for power to coil in motor contactor. If no voltage, check operation of reservoir level switch, high-temperature thermostat, cabinet access switch, flow switch and time delay relay. Electrically isolate each component and verify its open/close operation. Refer to Wiring Schematic, E&W Drawing T2583-1202-01. Replace, if necessary.
	c. Pump/motor malfunction	c. Check motor and pump, and replace, if necessary.
4. Excessive power draw	a. Short in wiring	a. Check wiring; repair or replace. Refer to Wiring Schematic, E&W Drawing T2583-1202-01.
5. Low GCWC coolant pressure	a. Pump/motor malfunction	a. Check motor and pump, and replace, if necessary.
	b. Low power to system	b. Check power and wiring; repair or replace.

TABLE IV-3
REPLACEMENT PARTS

PART DESCRIPTION	REFERENCE DESIGNATOR(1)	E&W PART NUMBER	QTY. PER GWHX
Contactor	K1	T2590-216	1
Controller with sensor, temperature	TC,TS	T2517-64	1
Pump/Motor set	P/M	T2583-101	1
Heat Exchanger	HX	T2590-105	1
Reservoir	R	T2590-104	1
Valve with actuator	BV,VA	T2583-120	1
Strainer	SI	T2583-124	1
Inlet, electrical, twist-lock	P/S	T2583-201	1
Display, temperature	TD	T2583-202	1
Transformer	T1	J1476-201	1
Switch, level	LS	T2583-205	1
Terminal block	TB1	29-28-182	6
Access valve	V1	T2583-122	1
Tubing, 1/2-inch (13-mm) inside diameter	--	T2583-103	10 ft. (3050 mm)
Tubing, 3/4-inch (19-mm) inside diameter	--	T2583-125	2.5 ft. (760 mm)
Thermostat, high-temp. limit	TH	T2590-106	1
Switch, interlock	IS	T2590-214	1
Switch, flow	FS	T2590-201	1
Relay, time delay	TDR	T2590-202,203	1
Light, power indicator	IL	T2590-215	1
Sensor, temperature	TS	T2517-64-01	1
Actuator, valve	VA	T2583-120-01	1
Cap, reservoir	CR	T2590-104-01	1
Assembly, hose, air purge	--	T2583-141	1
Cap, access valve	CA	T2583-122-01	1
Fuse block	F1	J1476-202	1
Fuse	F1	J1476-203	1
Assembly, quick-disconnect			
Facility chilled water			
Plug with hose barb	--	T2590-111	2
Socket with bulkhead and hose barb	--	T2590-112	2
Coil coolant			
Supply: Plug with bulkhead and	--	T2590-114	1

PART DESCRIPTION	REFERENCE DESIGNATOR(1)	E&W PART NUMBER	QTY. PER GWHX
hose barb	--		
Return: Plug with bulkhead and hose barb	--	T2590-121	1
Socket with hose barb		T2590-113	2

(1) Refer to E&W Drawings T2583-1201 and T2583-1202-01.

APPENDIX DRAWINGS

T2583-0101: INTERFACE CONTROL DRAWING, LTL-4
T2583-0201-01: UNIT ASSEMBLY, LTL-4
T2583-1201: PIPING SCHEMATIC, LTL-4
T2583-1202-01: WIRING SCHEMATIC, LTL-4

DWG NO T2583-1201 SH 1 REV D

NOTES:

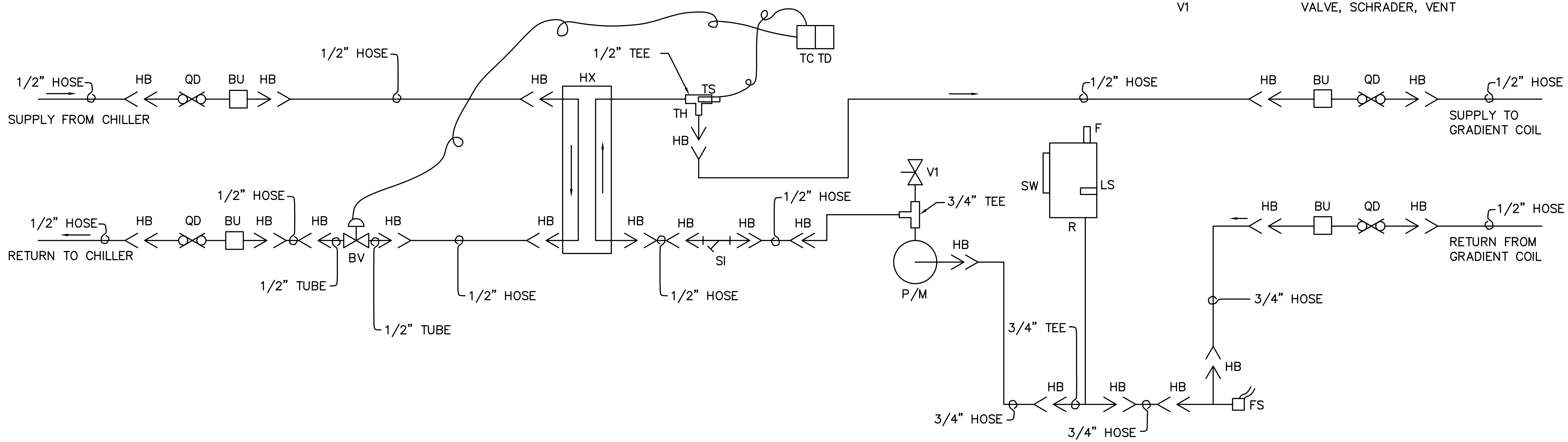
1. ALL HOSES ARE BRAID-REINFORCED PVC.

REVISIONS

ZONE	LTR	DESCRIPTION	DATE (YR-MO-DA)	BY	APPROVED	ENGINEERING APPROVED	Q.A. APPROVED
	A	REVISED PER ECN #T2583-01	98-10-15	MJS	BAA 98-10-20	KDD 98-10-20	RRS 98-10-20
	B	REVISED PER ECN #T2583-12	99-02-08	CMJ	BAA 99-02-08	KDD 99-02-08	RJR 99-02-08
	C	REVISED PER ECN #T2583-15	99-02-17	MJS	BAA 99-02-18	KDD 99-02-18	TJK 99-02-18
	D	REVISED PER ECN #T2590-03	99-05-25	MJS	BAA 99-05-25	KDD 99-05-25	TJK 99-05-25

LEGEND

BU	UNION, BULKHEAD
BV	BALL VALVE, 2-WAY
F	FILL CAP
FS	SWITCH, FLOW, COIL COOLANT
HB	HOSE BARB
HX	HEAT EXCHANGER, BRAZED PLATE
LS	SWITCH, LEVEL
P/M	PUMP/MOTOR SET
QD	QUICK DISCONNECT (WITH AUTOMATIC SHUT-OFF BOTH DIRECTIONS)
R	RESERVOIR
SI	STRAINER
SW	SIGHT WINDOW
TC	CONTROLLER, COIL SUPPLY TEMPERATURE
TD	DISPLAY, COIL SUPPLY TEMPERATURE
TH	THERMOSTAT, HIGH-TEMPERATURE LIMIT
TS	SENSOR, COIL SUPPLY TEMPERATURE
V1	VALVE, SCHRADER, VENT



SYM	NOMENCLATURE	CAGE CODE	IDENTIFYING NO.	MATERIAL / SPECIFICATION	UNIT WT.	ZONE	FIND NO.

PARTS LIST

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
TOLERANCES ARE
FRACTIONS ± N/A DECIMALS .XX = N/A .XXX = N/A ANGLES ± N/A
DO NOT SCALE DRAWING

CONTRACT NO.

DRAWN BY MJS DATE 98-09-30
CHECKER MJS 98-10-05 ENGINEER BAA 98-10-05
QUALITY ASSURANCE APPROVAL RRS 98-10-05
ENGINEERING APPROVAL KDD 98-10-05
MATL ENGR

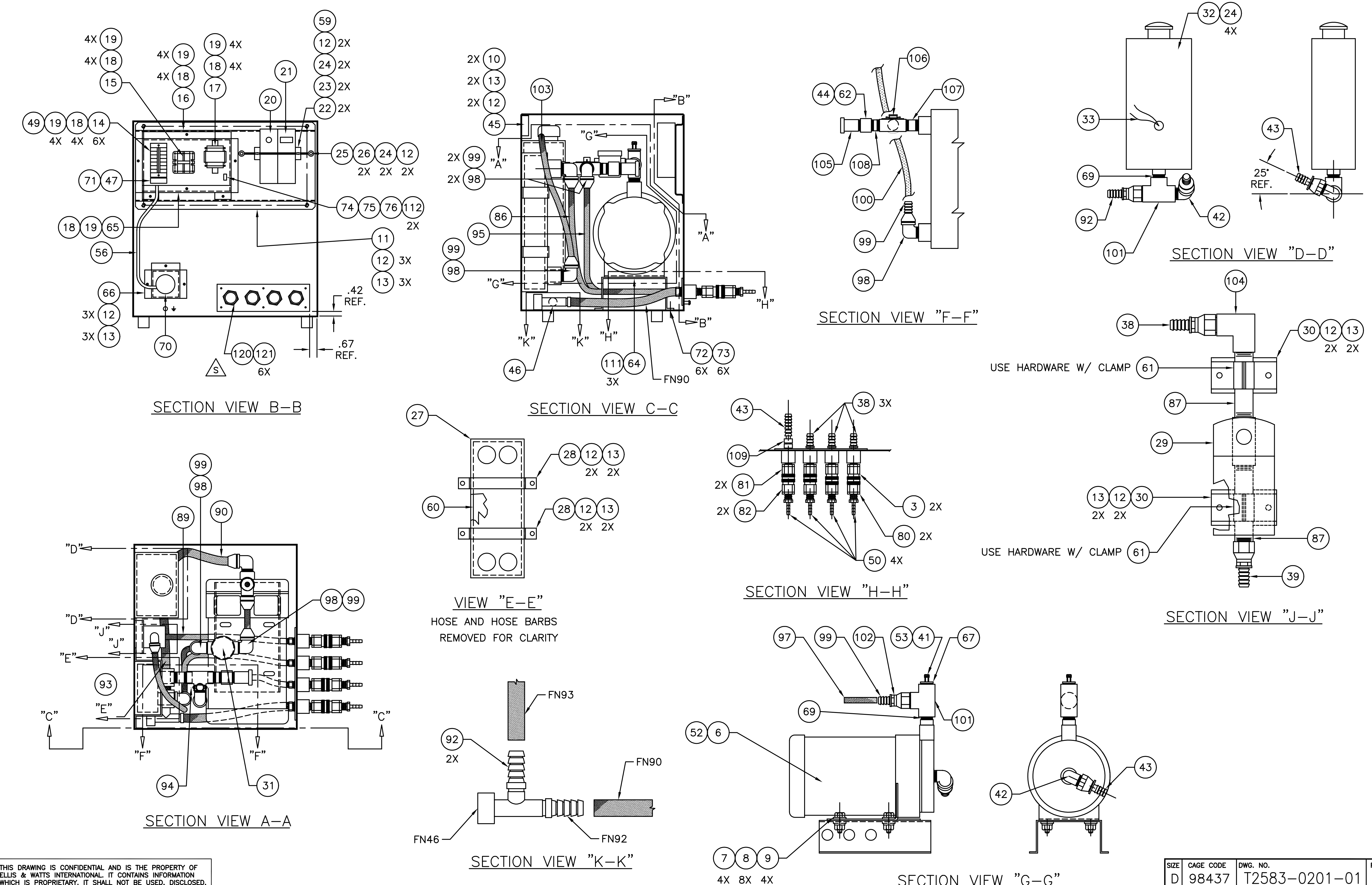
ELLIS & WATTS
CINCINNATI, OHIO

TITLE PIPING SCHEMATIC LTL-4

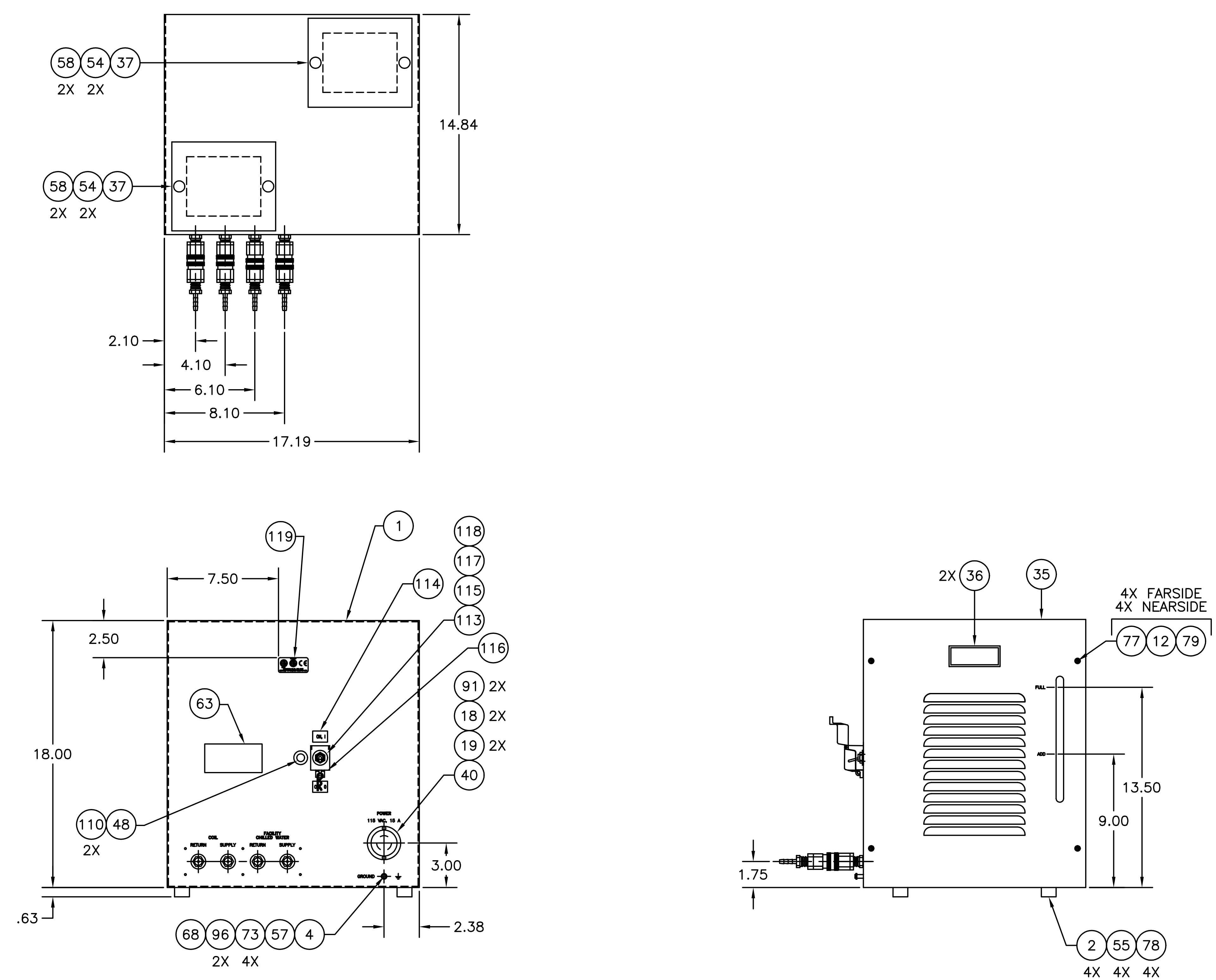
SIZE B CAGE CODE 98437 DWG. NO. T2583-1201 REV. D
SCALE NONE UNIT WT. SHEET 1 OF 1

PMIC	
T2583-0201-01	LTL-4
NEXT ASSY	USED ON
APPLICATION	

D	REV. No.	REV STATUS
1	SHEET No.	OF SHEETS



SIZE	CAGE CODE	DWG. NO.	REV.
D	98437	T2583-0201-01	S
SCALE 1/4	UNIT WT.	SHEET 3 OF 3	



-10 UNIT ASSEMBLY

SIZE	CAGE CODE	DWG. NO.	REV.
D	98437	T2583-0201-01	S
SCALE 1/4	UNIT WT.	SHEET 2 OF 3	

FILE: T2583-0201-01-REV-S 2

DWG NO	SH	REV
T2583-0201-01	1	S

REVISIONS

ZONE	LTR	DESCRIPTION	DATE (YR-MO-DA)	BY	APPROVED	ENGINEERING APPROVED	Q.A. APPROVED
	J	REVISED PER ECN# T2590-33,35,36,38-43	02-12-16	LG	MF 02-12-17 CAR 02-12-17	TJH 02-12-17	TJK 02-12-17
	K	REVISED PER ECN# T2590-44	03-02-04	JAM	LG 03-02-06 CAR 03-02-05	TJH 03-02-07	TJK 03-02-07
	L	REVISED PER ECN# T2590-46	03-09-30	MJS	LG 03-09-30 CAR 03-09-30	GAZ 03-10-01	RRS 03-10-01
	M	REVISED PER ECN# T2590-52	07-03-05	MAF	JAR 07-03-15 CAR 07-03-15	JKS 07-03-15	KDA 07-03-15
	N	REVISED PER ECN# T2590-56	07-03-22	MAF	MJS 07-03-22 JKS 07-03-22	TJH 07-03-22	RRS 07-03-22
	P	REVISED PER ECN# T2590-57	07-04-30	MAF	MJS 07-04-30 JKS 07-04-30	TJH 07-04-30	RRS 07-04-30
	R	REVISED PER ECN# T2590-58	07-05-04	MJS	MAF 05-04-07 CAR 05-04-07	JKS 05-04-07	RRS 05-04-07
	S	REVISED PER ECN# T2590-59,-60	07-08-15	MJS	JAD 08-15-07 JKS 08-15-07	PEB 08-15-07	RRS 08-15-07

NOTES:

1. REFERENCE DRAWING NO. T2583-0501 FOR LABELS.
2. ASSEMBLY PROCEDURES:
   - 2.1) HOSE BETWEEN RESERVOIR AND PUMP SUCTION.
   - 2.1.1) LOOSEN PUMP MOUNTING BOLTS.
   - 2.1.2) ROTATE ELBOW AT PUMP SUCTION TO 45° BELOW THE HORIZONTAL WITH HOSE BARB POINTING TOWARD BOTTOM OF RESERVOIR.
   - 2.1.3) CUT LENGTH OF 3/4-INCH I.D. HOSE.
   - 2.1.4) INSTALL HOSE BETWEEN RESERVOIR AND PUMP SUCTION, WHILE POSITIONING THE PUMP SO AS NOT TO STRESS THE RESERVOIR/TEE ASSEMBLY AND THE PUMP ELBOW.
   - 2.1.5) TIGHTEN THE PUMP MOUNTING BOLTS, TAKING CARE NOT TO STRESS THE RESERVOIR/TEE ASSEMBLY AND PUMP ELBOW.
   - 2.2) HOSE BETWEEN PUMP DISCHARGE AND STRAINER.
   - 2.2.1) CUT A LENGTH OF 1/2-INCH I.D. HOSE.
   - 2.2.2) STRAP STRAINER TO COPPER TEE USING TWO TYWRAPS.
   - 2.2.3) INSTALL HOSE BETWEEN TEE AT PUMP DISCHARGE AND STRAINER INLET. POSITION HOSE ON HOSE BARBS TO ELIMINATE STRESS AT THE COPPER TEE /HEAT EXCHANGER CONNECTION AND PVC TEE/PUMP DISCHARGE CONNECTION. TAKING CARE NOT TO STRESS THE PRECEDING CONNECTIONS.
   - 2.3) HOSE BETWEEN FLOW SWITCH AND RESERVOIR.
   - 2.3.1) INSTALL A BACKING JIG BETWEEN THE FLANGE ON THE CABINET BASE AND PVC TEE/HOSE BARB ELBOW ASSEMBLY AT THE BOTTOM OF THE RESERVOIR, TO PREVENT STRESSING OF THE RESERVOIR/TEE CONNECTION.
   - 2.3.2) INSTALL THE HOSE ON THE HOSE BARB AT THE RESERVOIR TEE.
3. PREPARE ALL METAL TO METAL THREADED CONNECTIONS PER ENG. PROCEDURE NO. 708, SECTION 2.0 AND ALL METAL TO PLASTIC AND PLASTIC TO PLASTIC THREADED CONNECTIONS PER ENG. PROCEDURE NO. 708, SECTION 4.0.

PARTS LIST

QTY	SYM	NOMENCLATURE	IDENTIFYING NO.	MATERIAL / SPECIFICATION	UNIT WT.	FIND NO.
6		POP-RIVET	50-18Z-114	1/8" DIA., ALUM.		121
1		ADAPTER MTG PLATE	T2583-0301-05			120
1		LABEL, ETL	J1476-102			119
1		PIN, COTTER, HAIRPIN	T2583-100D	1/4 TO 3/8 CLEVIS PIN DIA. 1/4 EYE DIA. .08 WIRE DIA. x 1-3/16"LG. 316 SS		118
1		WASHER, FLAT	50-3A-22	1/2"		117
1		LABEL OFF	SS1047-003			116
1		COVER, SWITCH GUARD	T2583-100C			115
1		LABEL, ON	SS1047-004			114
1		SWITCH	T2583-100B			113
2		CONNECTOR	SS2023-01			112
3		TERMINAL, RING	J1476-205			111
2		DISCONNECT, ELECTRICAL	T2590-206			110
1		CONNECTOR, MALE	J1476-209DZ	1/2" FLARE x 3/4		109
1		NIPPLE, PIPE	215PNL-8-15	1/2" NPT X 1 1/2" LG, MALE, BRASS		108
1		NIPPLE, PIPE	215PNL-8-20	1/2" NPT X 2" LG, MALE, BRASS		107
1		THERMOSTAT, SNAP-DISC.	T2590-106			106
1		COUPLING	62-1-320	1/2" FPT, BRASS		105
1		ELBOW, 90 DEG.	T2583-142	1/2" FPT, PVC		104
1		HOSE	SS1045-08-01600	1/2" DIA. x 16.00" LG.		103
1		MALE CONNECTOR	J1517-347	3/4"MPT x 1/2"MPT, FL. 37°, BRASS, PARKER#: 12-8 FTX-B		102
2		TEE	SS975-A4-12	3/4" x 3/4" x 3/4", PVC		101
1		HOSE	SS1045-08-00675	1/2" DIA. x 6.75" LG.		100
7		BARB, HOSE SWIVEL	J1476-211ZS	1/2"FPT x 1/2"HOSE, FEMALE JIC 37°, SWIVEL, ZPS, PARKER#: 30682-8-8B		99
6		ELBOW, 90 DEG.	J1476-210A	1/2" MPT x 1/2" MPT		98
1		HOSE	SS1045-08-00225	1/2" DIA. x 2.25" LG.		97
2		WASHER, LOCK, SST.	50-3E-3	1/4"		96
1		HOSE	SS1045-08-01725	1/2" DIA. x 17-1/4" LG.		95
1		SPRING	J1571-100			94
1		HOSE	SS1045-12-00575	3/4" DIA. x 5.75" LG.		93
3		BARB, HOSE STRAIGHT	J1476-209ZS	3/4"MPT x 3/4"HOSE, MALE NPTF PIPE, SWIVEL, ZPS, PARKER#: 31382-12-12		92
2		NUT, LOCK	50-2Z-151	#6-32, SST.		91
1		HOSE	SS1045-12-00550	3/4" DIA. x 5.50" LG.		90
1		HOSE	SS1045-08-01450	1/2" DIA. x 14.50" LG.		89
.4		LOCTITE SEALANT	J1476-215	LOCTITE NO MORE LEAKS		88
2		FITTING, NIPPLE	J1476-104	PVC, 1/2" MPT, 4" LG.		87
1		HOSE	SS1045-08-01800	1/2" DIA. x 18" LG.		86
.4		LOCTITE, PRIMER	21348	ACETONE		85
		ITEM DELETED				84

PARTS LIST

QTY	SYM	NOMENCLATURE	IDENTIFYING NO.	MATERIAL / SPECIFICATION	FIND NO.
.4		LOCTITE	56756	567 HI-TEMP SEALANT	83
2		DISCONNECT, QUICK, COUPLER	T2590-113ZS	EPR SEALS, BRASS	82
2		DISC., QUICK, NIPPLE	T2590-114-01ZS	EPR SEALS, BRASS	81
2		DISCONNECT, QUICK	T2590-111ZS	EPR SEALS, BRASS	80
8		WASHER, LOCK, SPLIT	50-3E-28	SST, #10	79
4		NUT, LOCK	MS51922-10	5/16-18, SST.	78
8		SCREW	50-19X-314	SST, #10-32 x 1/2	77
1	FI	SCREW, SELF TAPPING	J1476-101	SST, #2-56 x 1/2	76
1	FI	FUSE, CARTRIDGE	J1476-203	GLASS, SLO-BLO, 1A	75
1	FI	FUSEBLOCK	J1476-202	THERMOPLASTIC, BRASS, 10A, 300V	74
10		WASHER, FLAT	50-3E-34	SST, 1/4	73
6		NUT, LOCK	50-2Z-126	SST, 1/4-20	72
1		BRACKET	T2590-204	FOR TDR, T2590-202	71
2.5 IN.		TUBE, HEAT SHRINK	T2590-108	POLYOLEFIN, 2 DIA.	70
2		FITTING, NIPPLE	T2583-134	PVC, 3/4 MPT x 1.38 LG.	69
1		NUT, HEX	50-2Z-138	1/4-20, SST.	68
1		BUSHING	T2583-129	PVC, 3/4 MPT x 1/4 FPT	67
1		COVER, INLET, ELECTRIC	T2583-0301-03-15	LEXAN OR EQUAL, CLEAR, 0.06 THK.	66
1		RELAY COVER, BOTTOM	T2583-0301-03-13	LEXAN OR EQUAL, CLEAR, 0.06 THK.	65
1		PLATE, PUMP MOUNTING	T2583-0301-03-01	ALUM 5052-H32, 0.12 THK.	64
1		NAMEPLATE (LTL-4)	SS920-LTL-4	ALUM., 0.032	63
1		WELL, SENSOR, TEMPERATURE	SS650-001	1/2 MPT	62
2		CLAMP, UNISTRUT	SS477-2029	7/8 O.D.	61
2.4 FT²		INSULATION	SS302-7	FOAM RUBBER, ASTM-C-534, D-1056	60
1/2 FT		TRACK, MOUNTING	61M4-106		59
4		RIVNUT	50-2Z-155	PLATED STEEL, #10-32	58
1		BOLT	50-1A-4F	18-8 SST, 1/4-20 x 1 1/4, FULL THD.	57
A/R		TUBING, SPIRAL WRAP	29-37-278	NYLON, 1/4	56
4		WASHER, FLAT	MS15795-811	5/16" SST.	55
4		FASTENER, CAPTIVE SCREW	100-21-10	#10-32	54
1	CA	CAP, ACCESS VALVE	T2583-122-01	BRASS	53
1		ASSEMBLY, HOSE, AIR PURGE	T2583-141	PVC, BRASS	52
		ITEM DELETED			51
4		FITTING, HOSE BARB	62-1-191	BRASS, 1/2" H x 1/2" MPT	50
1		END STOP	29-28-183		49
1	IL	LIGHT, POWER INDICATOR	T2590-215	NEON, GREEN, 125 VAC	48
1	TDR	RELAY, TIME DELAY, W/POT	T2590-202,-203	120 VAC, 0.5 TO 60 SEC.	47
1	FS	SWITCH, FLOW	T2590-201	POLYPROPYLENE, NC>1 GPM, 20VA	46
1	IS	SWITCH, CABINET INTERLOCK	T2590-214	0.5A, 125 VAC	45
1	TC	THERMOSTAT, HIGH-TEMP. LIMIT	T2517-64		44
3		HOSE BARB, SWIVEL	J1476-208ZS	3/4"FPT x 3/4"HOSE, FEMALE JIC 37° SWIVEL, BRASS PARKER#: 30682-12-12B	43
2		ELBOW, 90 DEG.	J1476-207ZS	3/4"MPT x 3/4"MPT, FL. 37°, BRASS PARKER#: 12CTX-B	42

PARTS LIST

QTY	SYM	NOMENCLATURE	IDENTIFYING NO.	MATERIAL / SPECIFICATION	FIND NO.
1	VI	VALVE, ACCESS	T2583-122	1/4 MPT, HALF UNION	41
1	P/S	INLET, ELECTRICAL	T2583-201	TWIST-LOCK, 3 POLE, 15A, 125V	40
1		HOSE BARB	J1476-214	1/2" FPT X 1/2", ST.	39
4		HOSE, BARB	J1476-213ZS	1/2" HOSE x 1/2" MPT, ZPS PARKER# 31382-8-8	38
2		ACCESS PANEL	T2583-0301-03-03	ALUM 5052-H32, 0.080 THK.	37
2		HANDLE	T2583-112	PLASTIC	36
1		CABINET TOP	T2583-0301-02-01	ALUM 5052-H32, 0.080 THK.	35
A/R		GLYPTOL	270-079006-01		34
1	LS	SWITCH, LEVEL	T2583-205	POLYPROPYLENE, 20 VA	33
1	R	RESERVOIR	T2590-104	POLYEYHLENE, 1 GAL.	32
1	SI	STRAINER	T2583-124	POLYPROPYLENE HSG., 40 MESH SST SCREEN	31
2		UNISTRUT	T2590-116	ALUM.	30
1	BV, VA	VALVE, BALL, ACTUATED	J1476-204A	2 WAY, 1/2 NPT, 24 VAC	29
2		HX MOUNTING BRACKET	T2583-0301-03-05	ALUM 5052-H32, 0.12 THK.	28
1	HX	HEAT EXCHANGER	T2590-105	316 SST, BRAZED PLATE	27
2		RIVNUT	50-2Z-253	#10-32	26
1 FT.		SHOCKCORD	J1432-106	ø3/16"	25
8		SCREW, PHIL FL HD, SST	50-19X-342	#10-32 X .38 LG.	24
2		INSULATOR, TRACK MOUNT	61M4-110		23
2		MOUNTING CLIP	61M4-109		22
1	TD	TEMPERATURE DISPLAY	T2583-202	3 DIGIT	21
1	TC, TS	CONTROLLER, W/ TEMP. SENSOR	T2517-64	PROP + INTEG, -30 TO 130°F	20
19		WASHER, FLAT, SST	50-3E-25	#6	19
19		SCREW, PHIL FL HD, SST	50-19X-325	#6-32 X .50 LG.	18
1	TI	TRANSFORMER	J1476-201	115/230 VAC/12/24 VAC, 43 VA	17
1		RELAY COVER-TOP	T2583-0301-03-07	LEXAN OR EQUAL, CLEAR, 0.06 THK	16
1	K1	CONTACTOR	T2590-216	20A, 240/277VAC, 120 VAC COIL	15
6	TBI	TERMINAL BLOCK	29-28-182	6 POLE, 600V, 30A	14
16		NUT, HEX, LOCK	50-2Z-191	SST, #10-32	13
28		WASHER, FLAT, SST	50-3E-27	#10	12
1		MOUNT, ELECTRICAL	T2583-0301-03-11	ALUM. 5052-H32, 0.080 THK.	11
2		SCREW, PHIL FL. HD., SST.	50-19X-473	#10-32 x 1.25" LG.	10
4		NUT, LOCK	50-2Z-140	SST, 3/8-16	9
8		WASHER, FLAT	50-3E-36	SST, 3/8	8
4		BOLT, HEX HEAD	50-1A-36	SST, 3/8-16 x 1 1/4	7
1	P/M	PUMP/MOTOR	T2583-101	CENTRIFUGAL, 1 HP	6
		ITEM DELETED			5
1	GN	NUT, KNURLED THUMB, GROUND	T2583-113	304 SS	4
2		DISCONNECT, QUICK, COUPLER	T2590-112-02ZS	EPR SEALS, BRASS	3
4	VI	VIBRATION ISOLATOR	T2583-114	NATURAL RUBBER, STEEL BASE	2
1		HOUSING WELDMENT	T2583-0301-10	ALUM. 5052-H32, 0.080 THK.	1
–		UNIT ASSEMBLY	-10		–

PMIC	
T2583-0101	LTL-4
NEXT ASSY	USED ON
APPLICATION	

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
TOLERANCES ARE
FRACTIONS ± N/A; DECIMALS .XX = ±.06, .XXX = N/A; ANGLES ± N/A
DO NOT SCALE DRAWING

MATL ENGR

CONTRACT NO.

DRAWN BY CMJ — DATE 99-07-13
CHECKER MJS 99-07-14 — ENGINEER BAA 99-07-14
QUALITY ASSURANCE APPROVAL TJK 99-07-14
ENGINEERING APPROVAL MAD 99-07-14

ELLIS & WATTS INTERNATIONAL, INC. BATAVIA, OHIO

TITLE: UNIT ASSEMBLY (LTL-4)

SIZE	CAGE CODE	DWG. NO.	REV.
D	98437	T2583-0201-01	S

SCALE 1/4 — UNIT WT. — SHEET 1 OF 3

REV. No.	S	S	S
SHEET No.	3	2	1

REV STATUS OF SHEETS

DWG NO T2583-0101 SH 1 REV F

HEAT EXCHANGER — LTL SERIES

PERFORMANCE

COOLING CAPACITY: 4.5 KW, MAX.
COOLANT SUPPLY TEMPERATURE: 55° F TO 77°F
FREQUENCY: 50 HZ TO 60 HZ
ALTITUDE: 100 FT BELOW TO 8000 FT ABOVE SEA LEVEL
FLOW RATE: 2.5 GPM TO 3.0 GPM AT 50 HZ, 3.0 GPM TO 3.5 GPM AT 60 HZ
OUTLET PRESSURE: 20 PSIG TO 60 PSIG
COOLANT: 35% ETHYLENE GLYCOL/65%DEIONIZED WATER BY VOLUME
TEMPERATURE CONTROL: CHILLED COOLANT THROTTLING
RESERVOIR CAPACITY: 1 GAL.
COOLING FLUID: 50% ETHYLENE GLYCOL/50% WATER BY WEIGHT
MAXIMUM OPERATING AMBIENT: 113° F
MINIMUM OPERATING AMBIENT: 50° F

MECHANICAL

PUMP: CENTRIFUGAL, SINGLE STAGE, CPVC PLASTIC
COOLANT/COOLING FLUID EXCHANGER: BRAZED PLATE, SST
COOLANT CIRCUIT MATERIAL: PLASTIC, SST, COPPER
COOLING FLUID CIRCUIT MATERIAL: PLASTIC, SST, COPPER
PUMP INLET STRAINER: 40 MESH
RESERVOIR TYPE: POLYETHYLENE
MAXIMUM OUTLET PRESSURE: 60 PSIG
TEMPERATURE CONTROL VALVE: 2-WAY BALL
CABINET: ALUMINUM
METAL FINISHES: POWDER COAT
WEIGHT: 86 LBS. (DRY)

ELECTRICAL

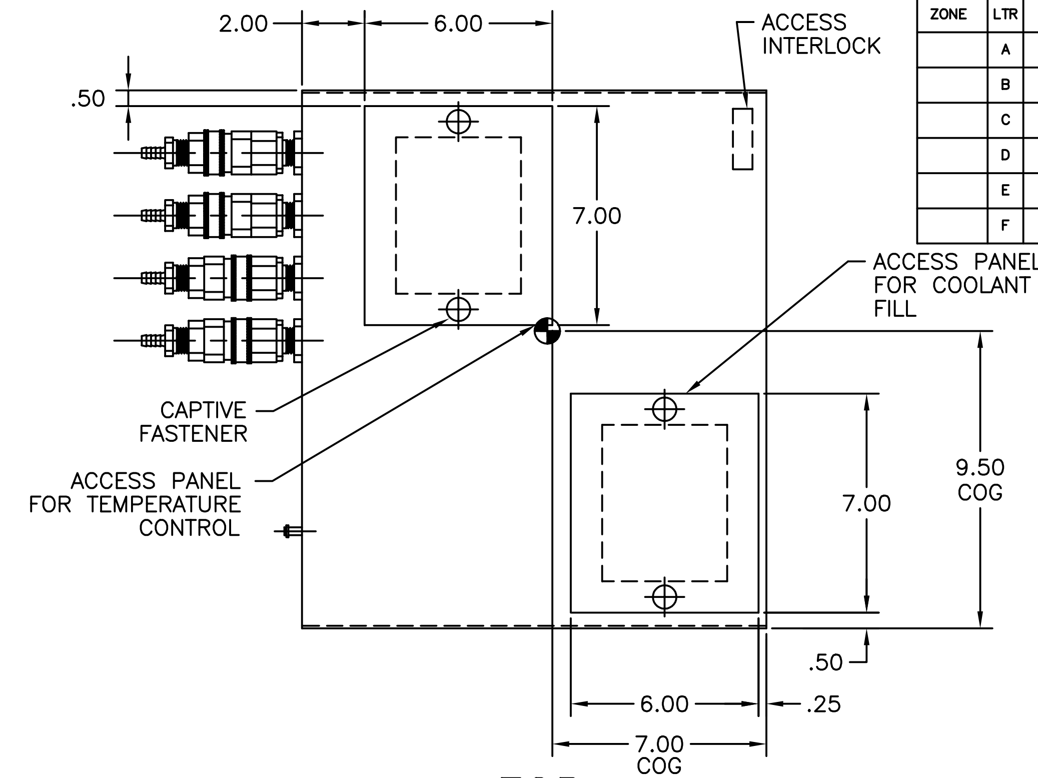
POWER SOURCE: 110VAC, 1PH, 50 TO 60 HZ
MAXIMUM POWER: 1000 VA APPROX.
PUMP MOTOR: 1.0 HP
MOTOR PROTECTION: THERMAL

FEATURES

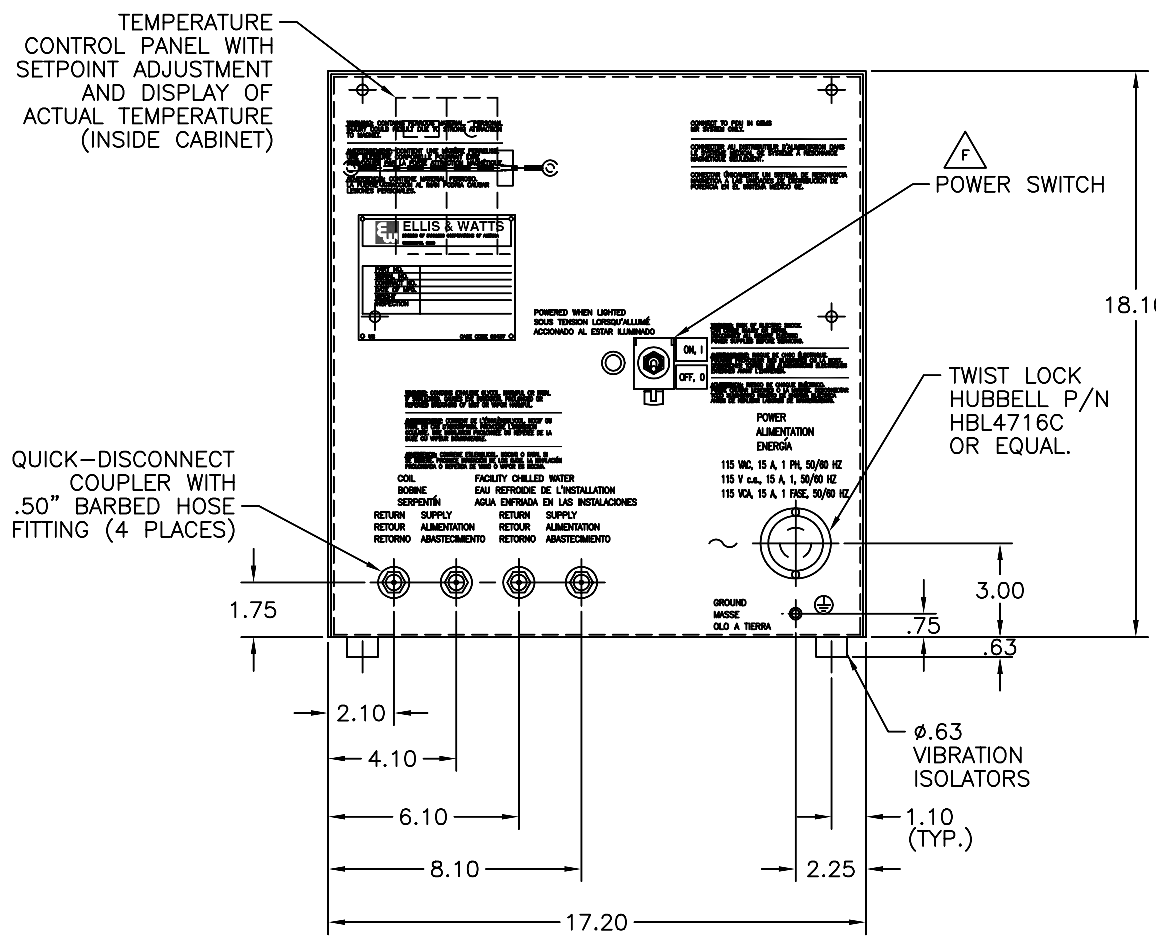
COOLANT SUPPLY TEMPERATURE CONTROL
COOLANT SUPPLY TEMPERATURE READOUT
QUICK-DISCONNECT FLUID COUPLERS WITH AUTOMATIC SHUT-OFF BOTH DIRECTIONS, POLARIZED COIL VS. CHILLER FLOW LOOPS
LOW COOLANT LEVEL INTERLOCK
LOW FLOW INTERLOCK
CABINET ACCESS SAFETY INTERLOCK
POWER INDICATOR LIGHT
HIGH-TEMPERATURE INTERLOCK

REVISIONS

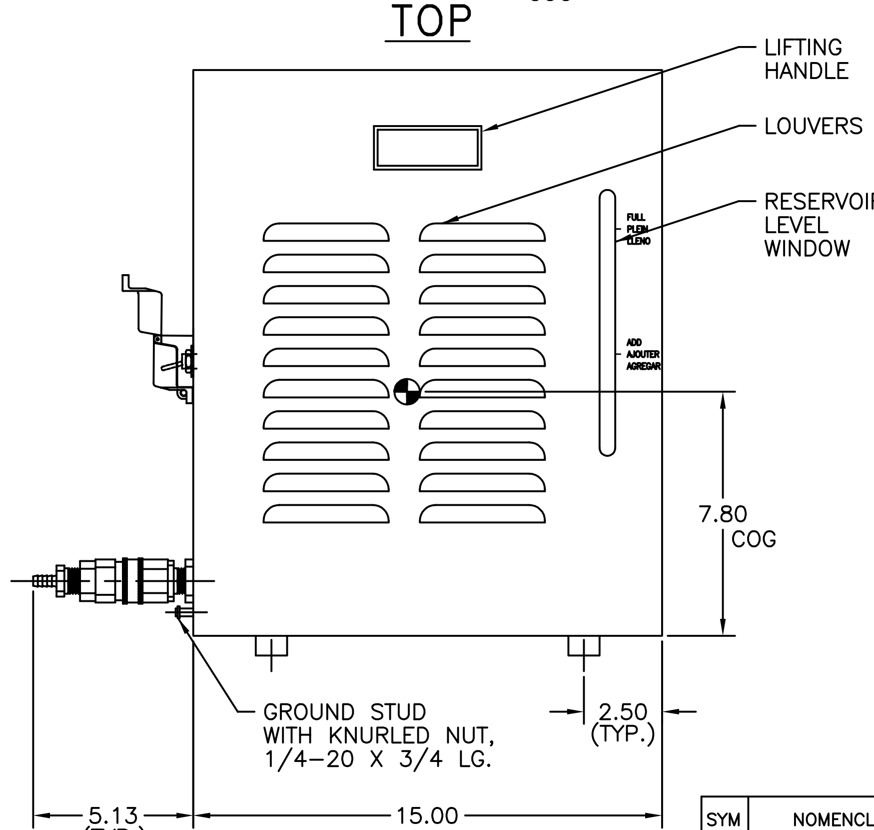
ZONE	LTR	DESCRIPTION	DATE (YR-MO-DA)	BY	APPROVED	ENGINEERING APPROVED	Q.A. APPROVED
	A	REVISED PER ECN #T2583-12	99-02-08	CMJ	BAA 99-02-08	KDD 99-02-08	RJR 99-02-08
	B	REVISED PER ECN #T2583-13	99-02-10	CMJ	BAA 99-02-11	KDD 99-02-11	RRS 99-02-11
	C	REVISED PER ECN #T2583-16	99-02-19	MJS	BAA 99-02-19	KDD 99-02-19	RRS 99-02-19
	D	REVISED PER ECN #T2590-04	99-04-29	CRL	BAA 99-05-26	KDD 99-05-26	TJK 99-05-26
	E	REVISED PER ECN #T2590-53	07-03-05	MAF	CAR 07-03-16	JKS 07-03-16	KDA 07-03-16
	F	REVISED PER ECN #T2590-54	07-03-21	MAF	JKS 07-03-21	TJH 07-03-21	RRS 07-03-21



TOP



LEFT SIDE



FRONT

SYM	NOMENCLATURE	CAGE CODE	IDENTIFYING NO.	MATERIAL / SPECIFICATION	UNIT WT.	ZONE	FIND NO.

PARTS LIST

PMIC	
	LTL-4
NEXT ASSY	USED ON
APPLICATION	

F	REV. No.	REV STATUS OF SHEETS
1	SHEET No.	

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
TOLERANCES ARE
FRACTIONS ± N/A — DECIMALS .XX = ±.13 .XXX = N/A — ANGLES ± N/A
DO NOT SCALE DRAWING

MATL ENGR

CONTRACT NO.

DRAWN BY CMJ — DATE 99-02-10
CHECKER MJS 98-11-10 — ENGINEER BAA 98-11-10
QUALITY ASSURANCE APPROVAL RJR 98-11-10
ENGINEERING APPROVAL MAD 98-11-10

ELLIS & WATTS
CINCINNATI, OHIO

TITLE: INTERFACE CONTROL DRAWING (LTL-4)

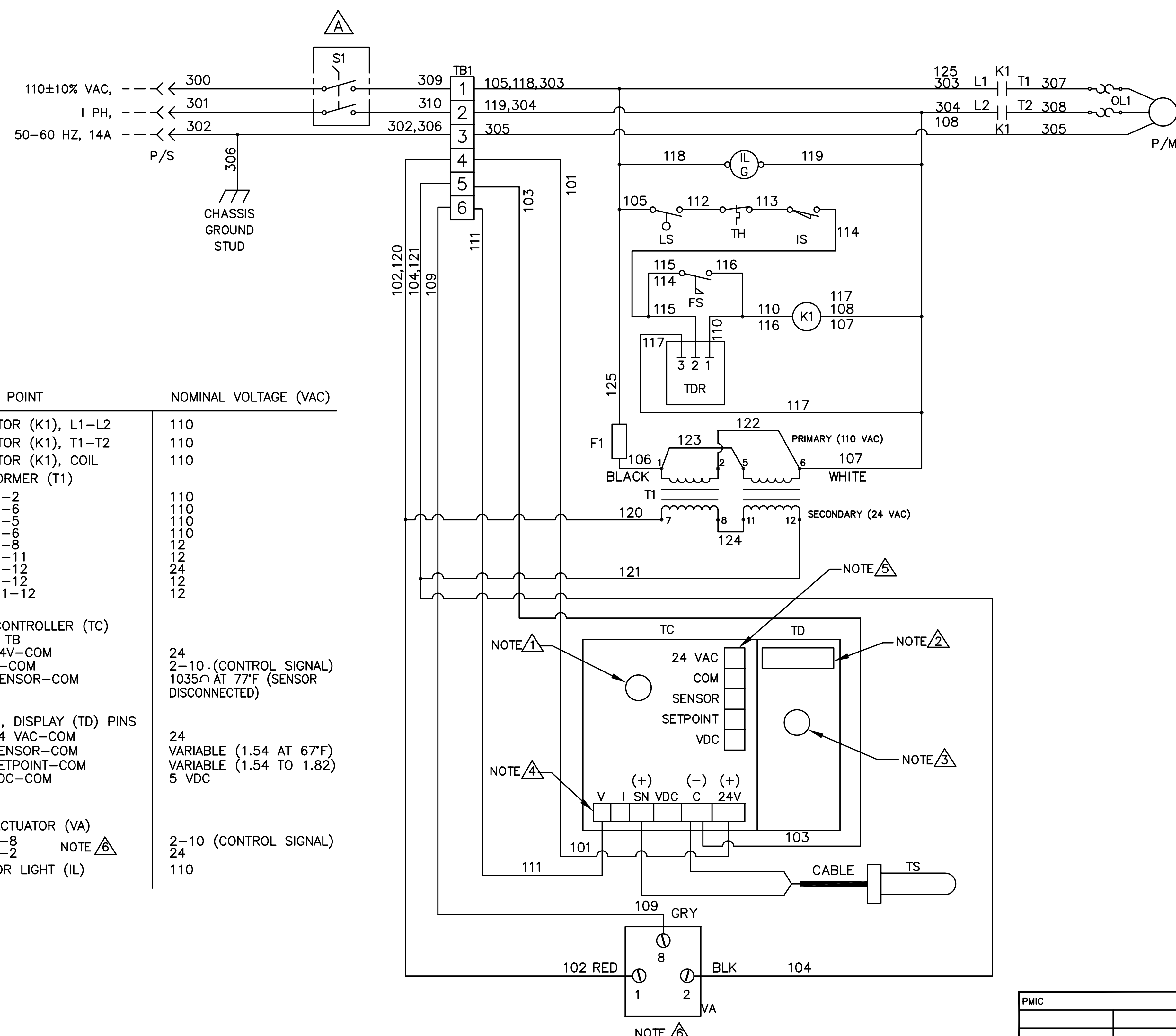
SIZE	CAGE CODE	DWG. NO.	REV.
D	98437	T2583-0101	F

SCALE 1/4 — UNIT WT. — SHEET 1 OF 1

FILE: T2583-0101-REV-F

REVISIONS							
ZONE	LTR	DESCRIPTION	DATE (YR-MO-DA)	BY	APPROVED	ENGINEERING APPROVED	Q.A. APPROVED
	A	SEE ECN-T2590-55 FOR CHANGES	07-03-22	MAF	MJS/CAR 07-03-22	JKS 07-03-22	RRS 07-03-22

DWG NO T2583-1202-01 SH 1 REV A



TEST POINT	NOMINAL VOLTAGE (VAC)
CONTACTOR (K1), L1–L2	110
CONTACTOR (K1), T1–T2	110
CONTACTOR (K1), COIL	110
TRANSFORMER (T1)	
1–2	110
1–6	110
2–5	110
5–6	110
7–8	12
7–11	12
7–12	24
8–12	12
11–12	12
TEMP. CONTROLLER (TC) MAIN TB	
24V–COM	24
V–COM	2–10 (CONTROL SIGNAL)
SENSOR–COM	1035Ω AT 77°F (SENSOR DISCONNECTED)
TEMP. DISPLAY (TD) PINS	
24 VAC–COM	24
SENSOR–COM	VARIABLE (1.54 AT 67°F)
SETPOINT–COM	VARIABLE (1.54 TO 1.82)
VDC–COM	5 VDC
VALVE ACTUATOR (VA)	
2–8	2–10 (CONTROL SIGNAL)
1–2 NOTE 6	24
INDICATOR LIGHT (IL)	110

LEGEND

F1	FUSE, 1A, 250V, SLO-BLO
FS	SWITCH, FLOW, COIL COOLANT, CLOSED FOR FLOW > 1 GPM
IL	LIGHT, INDICATOR, POWER (GREEN)
IS	SWITCH, INTERLOCK, CABINET ACCESS
K1	CONTACTOR, PUMP MOTOR, 20A
LS	SWITCH, LEVEL
OL1	OVERLOAD, MOTOR, THERMAL
P/M	PUMP/MOTOR SET, 1HP
P/S	PLUG/SOCKET, 3-POLE, 3-WIRE
A S1	SWITCH, ON/OFF
T1	TRANSFORMER, 110 VAC/24 VAC, 43 VA
TB1	TERMINAL BLOCK
TC	CONTROLLER, COIL SUPPLY TEMPERATURE, 3.2 VA CONSUMPTION
TD	DISPLAY, COIL SUPPLY TEMPERATURE
TDR	RELAY, TIME DELAY (30 SEC)
TH	THERMOSTAT, HIGH TEMPERATURE LIMIT, 100 ±7°F OPEN 80±7°F CLOSE
TS	SENSOR, COIL SUPPLY TEMPERATURE, SILICON
VA	ACTUATOR, 2-WAY BALL VALVE, 2VA CONSUMPTION

CONTROL LOGIC

1. TRANSDUCER (TS) SENSES ACTUAL COOLANT SUPPLY TEMPERATURE.
2. CONTROLLER (TC) COMPARES SET-POINT AND ACTUAL COOLANT SUPPLY TEMPERATURES.
3. USING PROPORTIONAL PLUS INTEGRAL LOGIC, CONTROLLER SUPPLIES 2-10 VDC SIGNAL TO ACTUATOR (VA) ON 2-WAY BALL VALVE.
4. TWO-WAY VALVE ADJUSTS TO PASS MORE CHILLED WATER THROUGH HEAT EXCHANGER, IF ACTUAL TEMPERATURE IS GREATER THAN SET-POINT TEMPERATURE AND LESS CHILLED WATER THROUGH HEAT EXCHANGER, IF ACTUAL TEMPERATURE IS LESS THAN SET-POINT TEMPERATURE.

NOTES:

1. KNOB TO DIAL-IN SET-POINT TEMPERATURE.
2. DISPLAY OF ACTUAL TEMPERATURE.
3. PRESS AND HOLD TO DISPLAY SET-POINT TEMPERATURE.
4. MAIN TERMINAL BOARD LOCATED INSIDE CONTROLLER.
5. ELECTRICAL PIN CONNECTIONS TO DISPLAY LOCATED INSIDE CONTROLLER.
6. THIS WIRING SCHEMATIC APPLIES TO UNITS WITH S/N'S 16806 AND GREATER. SEE DRAWING T2583-1202 FOR UNITS WITH S/N'S 16805 AND LESS.

WIRE DESIGNATION

A 101–125 = AWG #18
300–310 = AWG #12

SYM	NOMENCLATURE	CAGE CODE	IDENTIFYING NO.	MATERIAL / SPECIFICATION	UNIT WT.	ZONE	FIND NO.

PARTS LIST

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
TOLERANCES ARE
FRACTIONS ± N/A DECIMALS .XX = N/A .XXX = N/A ANGLES ± N/A
DO NOT SCALE DRAWING

PMIC

T2583-0201-01	LTL-4
NEXT ASSY	USED ON
APPLICATION	

MATL ENGR

CONTRACT NO.

DRAWN BY CMJ DATE 99-07-13
CHECKER MJS 99-07-13 ENGINEER BAA 99-07-13
QUALITY ASSURANCE APPROVAL RRS 99-07-13
ENGINEERING APPROVAL KDD 99-07-13

ELLIS & WATTS
CINCINNATI, OHIO

TITLE WIRING SCHEMATIC LTL-4

SIZE B CAGE CODE 98437 DWG. NO. T2583-1202-01 REV. A
SCALE NONE UNIT WT. SHEET 1 OF 1

			A	REV. No.	REV STATUS
			1	SHEET No.	OF SHEETS

FILE: T2583-1202-01-REV-A